JAMES WILLIAM PATTERSON

Manchester, United Kingdom

Email: james.patterson@gmail.com

Phone: +40 721 445 892

LinkedIn: linkedin.com/in/james-patterson-backend

GitHub: github.com/jwpatterson

PROFESSIONAL SUMMARY

Senior Backend Engineer with 8 years of hands-on experience building scalable distributed systems and leading technical initiatives across fintech and e-commerce sectors. I've spent the last four years mentoring junior developers while architecting microservices that handle millions of daily transactions. I'm most energised by solving complex database performance problems and enjoy working with teams who value both code quality and shipping velocity.

WORK EXPERIENCE

Senior Backend Engineer

TechVault Solutions Ltd

techvault.io

London, United Kingdom

March 2022 - Present

Designed and implemented a microservices architecture serving 50 million API requests daily, reducing latency by 40% through strategic caching and database query optimisation

Led the migration of a monolithic Python application to Golang, resulting in 60% improvement in system throughput and reduced infrastructure costs by 25%

Mentored a team of 5 junior backend engineers on design patterns, code review standards, and production debugging techniques, with 2 team members promoted to mid-level roles

Established comprehensive monitoring and alerting systems using Prometheus and ELK stack, reducing mean time to resolution for incidents from 45 minutes to 8 minutes

Backend Engineer

FinServe Digital

finserve.co

Manchester, United Kingdom

June 2019 - February 2022

Developed and maintained REST APIs in Java and Python handling real-time payment processing for 200+ enterprise clients with 99.95% uptime SLA compliance

Optimised PostgreSQL queries and implemented Redis caching strategies that reduced database load by 35% and improved API response times from 400ms to 120ms

Built a robust event-driven system using Apache Kafka to process transaction logs and feed data pipelines, enabling real-time fraud detection capabilities

Contributed to infrastructure automation using Terraform and Docker, enabling the team to reduce environment setup time from 2 days to 15 minutes

Software Engineer

DataStream Analytics Ltd

datastream.net

Manchester, United Kingdom

July 2017 - May 2019

Developed backend services in Python and Node.js for data ingestion and processing pipelines handling 5TB of data daily

Implemented comprehensive unit and integration test suites achieving 82% code coverage using pytest and Jest frameworks

Collaborated with frontend teams to design and document API contracts, reducing integration issues by 45%

Participated in on-call rotation and resolved production issues, becoming the go-to person for database performance troubleshooting

EDUCATION

Master of Science in Computer Science

University of Manchester

Manchester, United Kingdom

Graduated: June 2015

Bachelor of Science in Information Technology

University of Leeds

Leeds, United Kingdom

Graduated: June 2013

TECHNICAL SKILLS

Languages: Python, Java, Go, JavaScript, SQL, Bash

Databases: PostgreSQL, MySQL, Redis, MongoDB, Cassandra

Message Queues: Apache Kafka, RabbitMQ, AWS SQS

Cloud Platforms: AWS (EC2, RDS, Lambda, S3), Google Cloud Platform

Containerisation: Docker, Kubernetes, Docker Compose

Version Control: Git, GitHub, GitLab

Monitoring and Logging: Prometheus, ELK Stack, Grafana, DataDog

API Design: REST, GraphQL, Protocol Buffers

CI/CD: Jenkins, GitHub Actions, CircleCI

Architectural Patterns: Microservices, Event-Driven Architecture, CQRS, Saga Pattern

GITHUB REPOSITORIES

github.com/jwpatterson/payment-service-go - Production-ready payment processing service with comprehensive documentation and test coverage

github.com/jwpatterson/cache-layer-redis - Open-source Redis caching abstraction library used by 200+ projects

github.com/jwpatterson/kafka-stream-processor - Scalable data processing framework for Kafka event streams

github.com/jwpatterson/postgres-optimizer - Query analysis and optimisation tools for PostgreSQL performance tuning

CERTIFICATIONS

AWS Certified Solutions Architect Professional

Issued: September 2021

Certified Kubernetes Administrator

Issued: March 2020